

Yaqin Si

Raleigh, NC

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CORE SKILL SET

- Programming: Data cleaning, visualization, algorithm implementation, reproducible pipelines, and package development.
- Tools: R, Python, SAS, Julia, SQL, TreeAge, Tableau, GitHub, Shell scripts; Linux; HPC platform; AWS
- Statistics & Machine Learning: Advanced statistical modeling, simulation studies, experimental design, deep learning and large language models
- Data Expertise: genomics data (WGS, CNV/SV, SNP) and real-world EHR & Claims data
- Collaboration: Extensive experience working across academia, public health agencies, and industry
- Communication: Scientific writing, technical documentation, and presentations for both technical and non-technical audiences

EDUCATION

- **North Carolina State University** Raleigh, NC
Ph.D. Candidate in Bioinformatics – Advisors: Dr. Jung-Ying Tzeng & Dr. Wenbin Lu Aug 2020 – Aug 2026
Comajor: Statistics (GPA 3.9)
Relevant Coursework: Bioinformatics, Statistical Theory/Methods, Neural Networks, Functional genomics, Computation for Statistical Research, Computational Methods for Molecular Biology
- **Peking University** Beijing, China
Master in Epidemiology and Health Statistics, GPA 3.7/4.0 Aug 2016 – Jul 2018
Bachelor of Preventive Medicine & Bachelor of Economics, GPA 3.7/4.0 Aug 2011 – Jul 2016

EXPERIENCE

- **Research Assistant in Bioinformatics** Raleigh, NC
Bioinformatics Research Center, NC State University Aug 2020 – Present
CNV-profile regression with dual-penalties for DNA copy number variants (Method Development)
 - Developed **penalized regression** models using line-graph-guided penalties (**Weighted Fusion**) and **Lasso** to perform variable selection and smoothness of effect estimation of phenotype-associated CNVs.
 - Designed simulations to benchmark statistical power and estimation accuracy against existing methods.**The manuscript under revision at the *Annals of Applied Statistics***
R package development & Scalable Implementation of the above proposed methods
 - Released **CNVreg** R package on CRAN implementing genome-wide CNV-profile regression method.
 - Achieved **1,600+** downloads by the end of 2025, with full documentation and **reproducible vignette**.***Joint analysis of DNA structural variants and SNPs***
 - Built scalable computational pipelines to support joint analysis across CNV/SV and SNPs,
 - Integrated **haplotype imputation, heritability estimation, rare haplotype clustering, and polygenic risk score** construction to improve association power in whole-genome sequencing studies.
- **Data Scientist** Beijing, China
Beijing HealthCom Data Technology Company Jun 2018 – May 2019
 - **Established longitudinal cohort** by defining eligibility to perform epidemiological analysis of **incidence, prevalence**, daily hospital admissions, and direct medical costs by population subgroups.
 - Performed large data cleaning and feature engineering from **insurance claims** data using SQL on Hadoop.
 - Linked public **climate data** to build time-series models to forecast disease burden: incidence rate, comorbidity, and direct economic burden of diseases.

- o Practiced using **parsing, matching, and Natural Language Processing** to standardize unstructured diagnoses written by doctors and match to International Classification of Diseases 10 codes.
- o Built interactive Tableau dashboards to communicate insights to non-technical stakeholders.

Intern Research Assistant

Zhejiang, China

Jun 2015 – Mar 2016

- o **Emulated target trials** by defining eligibility, treatment assignment, and follow-up with integrated **electronic health records, insurance claims, and surveillance data** to support longitudinal analysis of incidence, prevalence, and mortality in different subgroups.
- o Built multivariate regression and survival models to identify risk factors and predict lifetime disease risk.
- o Conducted epidemiological **surveillance** for chronic disease management.

PROJECT

Bioinformatics Research Center, North Carolina State University

2020 – present

Hands on experience on causal DAG learning framework (Co-Lead)

Replicated and extended a published causal DAG learning framework (ANOCE) combining linear structural equation models (LSEM) with variational autoencoders (VAE)

Implemented constrained DAG structure learning with acyclicity and temporal ordering to estimate direct and indirect causal effects via mediators.

Evaluated model performance using TPR/FPR, F1-score, and Structural Hamming Distance (SHD) on simulated and real-world datasets.

Federated Learning for Privacy-Preserving and Efficient Distributed Modeling (A review) (Co-Lead)

Performed an in-depth literature review of federated learning frameworks, key challenges, and summarized current methodological solutions for heterogeneity-aware inference.

Terrain Identification from wearable device signals using Neural Networks (Co-Lead)

Implemented the Fully-connected Neural Network, CNN, and RNN with LSTM cell using Keras of TensorFlow; Tuned the network structure and hyperparameters based on validation loss and F1 score.

Naturally-inspired algorithm for multiple sequence alignment (Co-Lead)

Divided long sequences into more manageable pieces with Divide&Conquer strategy using genetic algorithm; Implemented Ant Colony Optimization algorithm for multiple DNA sequence alignment.

School of Public Health, Peking University

2016 – 2018

Disease prevention strategy evaluation based on risk prediction models

Conducted a systematic review and meta-analysis of statin effect for cardiovascular disease prevention (Lead); Built Cox Proportional Hazards Model for cardiovascular disease risk prediction based on Electronic Health Records database (Participate).

Evaluated public health intervention strategies using Markov Chain Monte Carlo simulations with public available evidence and summary statistics from cohort study to assist decision-making (Lead)

PUBLICATION (REFER TO GOOGLE SCHOLAR FOR A FULL LIST) AND DELIVERY

- o Si Y., Lu W., Tzeng J, et al, CNV-profile regression for CNV association analysis with whole genome sequencing data. **In revision at The Annals of Applied Statistics;**
Preprint available at bioRxiv (2024) <https://www.biorxiv.org/content/10.1101/2024.11.23.624994v1>
- o Si Y, Holloway S, Tzeng, J. CNVreg: An R package for CNV association analysis
CRAN <https://cran.r-project.org/web/packages/CNVreg/index.html>

ADDITIONAL INFORMATION

- o GGB symposium (2022) Best Student Presentation Award
- o Teaching assistant: Ph.D-level course ST721 Statistical Genetics (2023)